Appendix

A.Conversations with Advisor

Client:
I would like to build a study flow software, to be able to record the hours taken of revision for each subject on a sleek concise interface, enabling me to maintain the hours over the long term.

Advisor:
That's a great idea, but you need to remember that the IA is more so about the functionalities rather than the sleekness of the software. Now what are the inputs and outputs that you'd expect?

Client:
I would like there to be able to manually add the subjects and also have a timer that automatically saves the hours spent revising on a subject, where it can now keep a log of all sessions.

Advisor:
Okay that's great. This is already establishing the success criterias. Considering what you've said the core workflow of this software is the logging of sessions and the dynamic subjects creation.

Client:
I'll start with the planning of that then.

Figure 1: Conversation 1 with me and my advisor (Computer Science teacher)

Conversation 2: Development discoveries and technical touchups

Student (Client):
I have hit a roadblock. I literally just noticed that having a list of study sessions can make it hard for the interface to be understood to be interpreted. So I found out about the Recharts library whilst looking for data visualisation, and integrating charts can help ease interpretation.

Advisor:
This is quite a classic finding moment, but I'd like to know how charts can be integrated to the software in relation to the list of data to make correlation between the two of them.

Student (Client):
I would like to construct a trend line, as I've discovered that I could place a time series aggregation algorithm to place data into the Recharts library . I discovered that this is a lot more complex than expected.

Advisor:
This is good complex idea, this could be mentioned in Criterion C. Now this can update the expectations including a visual representation of the data. Showing that you can handle data transformation.

Student (Client):
I completely forgot to mention that I've noticed that the software loses data every time it refreshes. I want to add LocalStorage to be able to fix this problem

Advisor:
Good concern raised. Data persistence is essential to make sure that this software is fully functioning and stores the data so that when you refresh or leave and come back onto the page the data stays their.

Figure 2: Conversation 2 with me and my advisor (Computer Science teacher)

Conversation 3: Final Product and review

Student (Client):

The software is finally complete, I have cleaned up the whole software. I had so many difficulties, as my software is a desktop app which is running through npm. To load the desktop app i have to put the command “npm run dev,” in the terminal, but I kept receiving indications that there are many errors, so I had to go back and forth between different sources to understand errors made

Advisor:

Okay well that is interesting, so what type of difficulties did you face during the process

Student (Client):

Firstly, I had made plenty of typo mistakes, which vscode helped me by indicating where they were and I would go over and input the correct format of certain lines. Secondly, an issue of dependencies in the package lock json, which stemmed from the package json. Lastly, the colour scheme of the website irritated me so much, where I wanted the desktop software to look aesthetically pleasing, based on one of my success criterions.

Advisor:

So now I understand more of the issues and what you do to solve those problems that you faced.

Student (Client):

Well first things first I tried my best to do my own independent research in terms of both situations of the incorrect wording for the code and the dependencies issues, now for the incorrect wording, i was able to make the fixtures independently. For the dependencies, I had asked for aid from you as my advisor which the problem was automatically fixed. Lastly, the package lock json that one was difficult because the issue was truly from the package json so I had asked my classmate to guide me through the fixing process and that is practically it, this was a long but fulfilling process.

Advisor:

This is a great milestone that you've faced and do you think that there are any improvements you believe you can make or any other concerns.

Student:

Well I will say that this software desktop could have more functionalities such as a scoreboard if it would have been a public website so that students can compete between each other, but I will say because this was for one individual I did just want to limit the desktop software to fulfill the specific criterions and i also do not have any other concerns.

Figure 3: Conversation 3 with me and my advisor (Computer Science teacher)